

Project Proposal

Jorge Garcia

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For my project, I plan to build a WebGPU-based renderer for visualizing Constructive Solid Geometry (CSG) scenes in real time. My goal is to implement a GPU pipeline that can evaluate these operations dynamically, without relying on static vertices. This will allow the system to render geometry that is defined procedurally at runtime. To make the system easy to use, I will design a small domain-specific language that describes CSG scenes in a concise, readable way. The DSL will support declaring primitives, applying transformations, and combining objects through CSG operations.

A major learning objective of this project is to explore “dynamic” graphics programming, where the geometry is not defined ahead of time as static vertex arrays but instead evaluated procedurally on the GPU. It also includes techniques such as raymarching or other procedural methods that allow the GPU to compute intersections, apply transformations, and evaluate boolean geometry trees on the fly. I also want to learn more on how WebGPU works and how it differs from WebGL and OpenGL.

The final artifact will be an interactive browser renderer that takes a scene description written in the DSL, compiles it into an intermediate representation, and renders the resulting CSG object using WebGPU. If time permits, I may extend the system with additional features such as mouse-draggable camera controls, more lighting options, and/or material coloring.